A Non-Absolute Schauder Frame in a Nuclear Frechet Space

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2026-07-03

Source Question

This packet concerns Remark 2.6 of Cassese–Ramos, *On the regularity of Fourier interpolation formulas*, arXiv:2503.15733. The paper defines a Schauder frame in a locally convex space X as a sequence $((x_n, y_n))_n$, with $x_n \in X$ and $y_n \in X'$, such that

$$x = \sum_{n=1}^{\infty} y_n(x) x_n$$

for every $x \in X$, with convergence in the topology of X. Immediately after this definition, the paper asks whether, in view of the corresponding theorem for bases, a Schauder frame in a nuclear Frechet space must be absolute.

0 otherwise

It is easy to see that (x_n, y_n) is a Schauder frame and $\sum c_n x_n$ converges, hence $\{y_n\}$ is a strong $\Lambda-$frame. ■

Remark 2.6. The results above hold, more generally, if Λ is a barrelled locally convex sequence space with $\{e_n\}$ a Schauder basis. It is natural to wonder, given the results on bases, if a Schauder frame in a nuclear Fréchet space must be an absolute Schauder frame. As it turns out, this problem is currently open.

To give the reader an idea of the complexity of the problem, let us mention a slightly more general version that is known to be false: a sequence $\{x_n\} \subset X$ is said to be a *representing system* in X if for each $x \in X$ there exists a sequence (α_n) such that

$$x = \sum \alpha_n x_n.$$

It is easy to see that the concept is a generalisation of both Schauder bases and frames; in this setting, it is not true that every representing system in a Fréchet nuclear space is an absolute system, as shown by Kadets and Korobeinik in [32].

2.2. **Fredholm Operators and sc-Fredholm Operators.** We next lay out the foundation for the application of the theory of sc-Fredholm operators, as mentioned in the introduction. We first recall the definition of semi-Fredholm and Fredholm operators:

Definition 2.7. Let X, Y be locally convex spaces. An operator $T : X \to Y$ is said to be upper (respectively, lower) semi-Fredholm if its range is closed and its kernel finite-dimensional

We interpret "absolute Schauder frame" in the direct frame analogue of the source paper's Definition 2.1 for absolute bases: for every $x \in X$ and every continuous seminorm p, the scalar series

$$\sum_{n=1}^{\infty} |y_n(x)|\, p(x_n)$$

converges. The counterexample below also fails the stronger bounded-set version obtained by replacing $|y_n(x)|$ with $p_B(y_n) = \sup_{x\in \mathrm{co}(B)} |y_n(x)|$, by taking $B = \{e_1\}$.

Answer

The answer is negative. There is a Schauder frame in the nuclear Frechet space s of rapidly decreasing scalar sequences whose reconstruction series converges in s for every vector but is not absolute even at one basis vector.

Proof Intuition

Start with the usual absolute basis (e_j) of s. Replace each single basis coefficient by a full conditionally convergent scalar row: choose scalars r_k with $\sum_k r_k = 1$ but $\sum_k |r_k| = \infty$. The j-th coordinate is then reconstructed as $a_j \sum_k r_k e_j = a_j e_j$. Enumerating the double sequence by diagonals makes every fixed coordinate see longer and longer initial sums of $\sum r_k$, while the partial sums remain uniformly bounded. Rapid decay of a_j controls the moving tail over j. Thus the frame converges in s, but absolute convergence fails on e_1 because it contains the divergent series $\sum_k |r_k|$.

The Counterexample

Let

$$s = \left\{ a = (a_j)_{j\geq 1} : \ p_m(a) := \sup_{j\geq 1} j^m |a_j| < \infty \text{ for every } m \in \mathbb{N}_0 \right\}.$$

This is the usual Frechet space of rapidly decreasing sequences. It is nuclear: indeed, for $m \in \mathbb{N}_0$ choose $\ell > m+1$. The canonical map from the Banach completion for p_ℓ to the Banach completion for p_m has the representation

$$a \mapsto \sum_{j=1}^{\infty} e_j^*(a) e_j,$$

where $\|e_j^*\|_{p_\ell} \leq j^{-\ell}$ and $p_m(e_j) = j^m$, so

$$\sum_{j=1}^{\infty} \|e_j^*\|_{p_\ell}\, p_m(e_j) \leq \sum_{j=1}^{\infty} j^{m-\ell} < \infty.$$

This gives the standard nuclearity criterion for s.

Set

$$r_k = \frac{(-1)^{k+1}}{k \log 2}, \qquad k \geq 1.$$

Then $\sum_{k=1}^{\infty} r_k = 1$, $\sum_k |r_k| = \infty$, and the partial sums $\sigma_K = \sum_{k=1}^{K} r_k$, with $\sigma_0 = 0$, are bounded. Let

$$C_0 = \sup_{K\geq 0} |\sigma_K| < \infty.$$

Enumerate $\mathbb{N}\times\mathbb{N}$ by diagonals, in increasing order of $j+k$, and within each diagonal in increasing order of j. Write the resulting enumeration as

$$n \mapsto (j_n, k_n).$$

Define

$$x_n = e_{j_n} \in s, \qquad y_n = r_{k_n} e^*_{j_n} \in s',$$

where $e^*_j(a) = a_j$. Each y_n is continuous.

Theorem 1. *The sequence $((x_n, y_n))_{n\geq 1}$ is a Schauder frame for the nuclear Frechet space s, but it is not an absolute Schauder frame.*

Proof. Fix $a = (a_j) \in s$, and let

$$S_N a = \sum_{n=1}^{N} y_n(a) x_n.$$

Because of the diagonal ordering, for each j the terms from the j-th row appear in their natural order $k = 1, 2, \ldots$. Hence

$$S_N a = \sum_{j=1}^{\infty} a_j \sigma_{K_j(N)} e_j,$$

where $K_j(N) \in \mathbb{N}_0$, $K_j(N) \to \infty$ for each fixed j, and

$$\sup_{j,N} |\sigma_{K_j(N)}| \leq C_0.$$

The displayed sum is finite for each N, but this coordinate formula is the convenient way to compare it with a.

We prove $S_N a \to a$ in every seminorm p_m. Let $\eta > 0$. Since $a \in s$, choose J so large that

$$\sup_{j>J} j^m |a_j| < \frac{\eta}{2(C_0+1)}.$$

For the finitely many indices $1 \leq j \leq J$, the convergence $\sigma_{K_j(N)} \to 1$ implies that, for all sufficiently large N,

$$\max_{1\leq j\leq J} j^m |a_j|\, |\sigma_{K_j(N)} - 1| < \frac{\eta}{2}.$$

For these N,

$$\begin{aligned} p_m(S_N a - a) &= \sup_{j\geq 1} j^m |a_j|\, |\sigma_{K_j(N)} - 1| \\ &\leq \max_{1\leq j\leq J} j^m |a_j|\, |\sigma_{K_j(N)} - 1| + (C_0+1) \sup_{j>J} j^m |a_j| \\ &< \eta. \end{aligned}$$

Thus $S_N a \to a$ in s, so $((x_n, y_n))$ is a Schauder frame.

It is not absolute. Take $a = e_1$ and $p_0(a) = \sup_j |a_j|$. The terms with $j_n = 1$ contribute

$$|y_n(e_1)|\, p_0(x_n) = |r_{k_n}|\, p_0(e_1) = |r_{k_n}|,$$

and therefore

$$\sum_{n=1}^{\infty} |y_n(e_1)|\, p_0(x_n) \geq \sum_{k=1}^{\infty} |r_k| = \infty.$$

Consequently the frame is not absolute. The same computation with $B = \{e_1\}$ shows failure of the stronger bounded-set absolute condition as well. $\square$

Verification Notes

The proof is purely analytical. The only external source used for the target statement and definitions is the Cassese–Ramos paper. A finite numerical sanity check in `code/check_diagonal_frame.py` illustrates the diagonal partial-sum mechanism and the growth of $\sum_{k\leq N} |r_k|$; it is not used as proof.

Novelty Check

Before promotion, the run indexes were searched for close variants of

`Schauder frame`, `nuclear Frechet`, `absolute Schauder frame`, `representing system`.

No prior packet or attempt answered this question. Web searches were also run for close phrases including "Schauder frame absolute nuclear Frechet", "Schauder frames nuclear Frechet absolute", and "absolute Schauder frame". No prior answer was found in the bounded search. Novelty confidence is therefore moderate pending expert review.

References

References

[1] G. Cassese and J. P. G. Ramos, *On the regularity of Fourier interpolation formulas*, arXiv:2503.15733, 2025.